

VMD MIC Solution

Big picture take-aways and questions (not required):

- Cost system inform strategic decisions such as pricing, product mix, and investments in technology.
 - At a high level, the case is about a simple (and common) problem: when margins decline, is it due to prices being too low or costs being too high?
- The company prices their products at full cost. Yet we know that prices in the market are determined by demand and supply and market is at least approximately competitive. How are these connected? If our cost data do not adequately capture market economics, we will get pricing wrong and lead to a “demand-driven deal spiral”.
- Should employees/managers be held accountable for costs they do not control? (For example, in this case, should we allocate costs like landscaping to the products?) Related, if we want to use costs in performance evaluation, we need to be sure that those being evaluated understand and agree with the way this is done – participation and buy-in is critical in using costs for decision-making (such as pricing) and to evaluate employees (as at least implicitly occurs in the case – Jankoski reacts defensively to the implication that operations are not cost efficient).
- Could they implement a contribution margin approach work here? (Perhaps in the short-run but in the long-run the entity needs to recover all costs.)
- The issues identified in this case are analogous to those in the Pen Factory example and help motivate how ABC can be helpful.
- Related to the ABC framework, we may want to think about whether we are thinking about the right cost objects. The case focuses on the four types of tests (the products) but perhaps we could think about this in other ways – for example, should we think about the lifetime value of the WNE relationship? Does it make sense to think about the products sold to them separately?

Q1.

The burden rate has increased from 93% five years ago to 144% (top of page 3), which suggests overhead costs are becoming more important relative to direct labor. In general, an increasing burden rate, especially a rate that exceeds 100%, suggests overhead costs have become relatively more important and so can indicate that the underlying activity measure/cost driver (here direct labor costs) may no longer be a good measure of what drives overhead costs.

As described in the case (top of page 3), overhead costs include:

- Maintenance of physical facility
- General testing of equipment and supplies

- Energy and water costs
- Year-round landscaping (mowing, gardening, snow removal)
- Outsourced, specialized maintenance of equipment required to comply with regulatory standards.

It seems clear from this that a large variety of costs are included in this single overhead cost pool, and it is hard to see how all are likely to be related to a single overhead cost driver such as direct labor \$ for the four products. For example, while maintenance of the equipment may relate to usage as measured by direct labor hours, it is hard to see how energy/water costs (or landscaping) are related to direct labor costs.

This helps explain why Mr. Jankoski is unhappy – because he manages operations his responsibilities include operating the equipment used for the four products but not the overall costs of running the facility, the costs of water and energy, landscaping, etc., and yet he is being held responsible for the higher overhead costs (and so the pricing) of the products.

We may want to consider whether it makes sense to allocate the cost of activities like landscaping and water (which seem unrelated to the provision of test services to customers) to the products at all. Perhaps they are best thought of as fixed costs of the business overall and managed separately. This would make Jankoski a bit happier (because these costs for which he is held responsible would only include those that he can control – i.e., those resources over which he has decision rights).

Q2.¹

See spreadsheet for calculations. What are the pros and cons of the simple system (allocate all overhead based on DL \$) compared to the more complex systems that use two or three cost pools?

- The basic tradeoff is simplicity (of calculation and information) versus greater “accuracy” of costing that comes at a price—as we add more cost pools, the relation between the activity driver and costs in each pool becomes stronger (more linear). The case provides little information about the cost of collecting data on machine run times (separately for high tech and low tech under the Posh system) but this could be significant.
- Notice that the allocations we end up with under the 2-cost pool and 3-cost pool systems are similar, leading to questions about whether the more complex Posh system would meet the cost-benefit test relative to that proposed by Crest (both seem like improvements relative to the use of a single cost pool).

¹ In general, my suggested solutions contain more information and commentary than you are expected to provide.

What's most accurate?

Technically, the “most accurate” system is the 3-pool system as implemented by Posh – in the sense that with more cost pools there is a stronger relationship between the activity measures used as the basis for cost allocation (the cost drivers) and the costs in each pool. For example, presumably “high tech hours” is more closely related to high tech overhead costs than direct labor costs.

Which to use for pricing?

As stated in the case, the prices that VMD charges for its four products are based on the full cost of each (i.e., after allocating all overhead costs).

As a result, because VMD uses DL \$ to allocate overhead and overhead is becoming more important overall (because of the increasing importance of the higher tech products), VMD overprices services with a relatively high DL component (x-rays and fluoroscopies) and underprices services for which the opposite is true (CAT scans and MRIs, especially high-end 3D MRIs).

[This assumes that external market prices for the products are about right (that the market is competitive), which implies our competitors are doing a better job in pricing these services. This will be especially true for competitors that just offer x-rays and fluoroscopies; these competitors will have significantly lower overhead costs because they are not supporting CAT scans and MRIs.]

Given this mispricing, we will lose customers for products with relatively high DL \$ (x-rays and fluoroscopies) which we overprice, and gain customers – but ultimately lose money – for products with relatively low DL \$ (CAT scans and MRIs), which we underprice.

This leads to a *demand-induced death spiral*: as this continues, our sales mix moves towards more machine-intensive products, which in turn leads to a further increase in machine (overhead) costs relative to DL costs. This will cause a further increase in the burden rate (numerator increases while denominator declines) which exacerbates the problem until we are eventually completely priced out of the market for the high DL \$ products (x-Rays and fluoroscopies) and price below cost for the higher tech products, which will eventually put us out of business because we can't cover our costs.

Under what circumstances would a simple approach work best? This approach would work in industries where DL are relatively important (in the context of all costs) and where product lines are well-established, homogeneous, and stable. This is *not* the case for VMD, where products are increasingly heterogeneous and the market is shifting towards the use of technology-intensive products, such that the importance of DL costs is falling relative to overhead costs, especially for the high-tech equipment.

Q3.

As indicated in the question, depreciation is normally an overhead cost and so allocated to the product along with the other overhead costs. Naturally, depreciation of a given piece of equipment would be a fixed cost, which raises the question of the right activity measure.

Although depreciation is a sunk cost, it nevertheless reflects a real economic cost incurred by the business (CAPEX). For a business to survive in the long run it needs to recover all costs, including CAPEX. However, for short-run pricing, a contribution margin approach may be more appropriate when thinking about pricing (i.e., if we're already committed to having the machine in service for a given period, we may consider pricing to recover our short run variable costs, keeping in mind that this is not sustainable in the longer run).

Q4.

Conceptually, this is a good idea because all of the costs and benefits of operating, marketing, selling, and maintaining this new equipment will be fully internalized by Olive. The problem arises if there are interdependencies between the decisions that she makes and the rest of our business (if there is some interdependency between the costs (production functions) and/or the demand functions of the 3D MRI machine and the 2D MRI machine and other products). For example, if there is some substitutability between 2D and 3D MRI products and she prices 3D MRI above or below market; if customers want a seamless experience prefer to deal with a single entity and we effectively now operate as two entities neither of which provides the full range of services; if we could be saving some fixed costs by operating all equipment as part of same operation (e.g., Olive decides to hire her own landscape crew to try and differentiate her product). Perhaps more important, if the new business unit shares the facilities, how do we allocate the shared (overhead) costs, such as landscaping and utilities, which brings us back to the same allocation problem.

Q2. Product costing under the three different allocation systems.

Existing
System:

	DL \$	Allocated OH	Full cost	Share
X-Rays	417,073	601,072	1,018,145	0.303
Fluoroscopies	414,047	596,711	1,010,758	0.301
CAT Scans	270,850	390,340	661,190	0.197
MRI Scans	273,601	394,305	667,906	0.199
	1,375,571	1,982,428	3,357,999	1.000
Total overhead	1,982,428			
Allocation Rate	1.4412			

System
proposed by
Crest:

	Run time (hours)	DL \$	Allocated DL Overhead	Allocated Equip OH	Total OH Allocated	Full cost	Share
X-Rays	1,060	417,073	218,726	212,816	431,542	848,615	0.253
Fluoroscopies	1,312	414,047	217,139	263,410	480,549	894,596	0.266
CAT Scans	1,394	270,850	142,042	279,873	421,915	692,765	0.206
MRI Scans	2,515	273,601	143,485	504,936	648,421	922,022	0.275
	6,281	1,375,571	721,392	1,261,036	1,982,428	3,357,999	1.000

Allocation rate: DL-related overhead

Overhead	721,392
Direct Labor \$	1,375,571
Rate	0.52

Allocation rate: Equipment-related overhead

Overhead	1,261,036
Total hours	6,281
Rate	200.77

System
proposed by
Posh:

	Lo-tech hrs.	Hi-tech hrs.	DL \$	Allocated DL Overhead	Allocated Equip OH Low Tech	Allocated Equip OH High Tech	Total OH Allocated	Total cost	Share
X-Rays	792	268	417,073	218,726	127,943	59,086	405,756	822,829	0.245
Fluoroscopies	733	579	414,047	217,139	118,412	127,653	463,204	877,251	0.261
CAT Scans	575	819	270,850	142,042	92,888	180,566	415,497	686,347	0.204
MRI Scans	0	2,515	273,601	143,485	0	554,487	697,972	971,573	0.289
	2,100	4,181	1,375,571	721,392	339,243	921,793	1,982,428	3,357,999	1.000

Allocation rate: DL-related overhead

Overhead	721,392
Direct Labor	1,375,571
Rate	0.52

Allocation rate: High Tech Equipment-related overhead

Overhead	921,793
Total hours	4,181
Rate	220.47

Allocation rate: Low Tech Equipment-related overhead

Overhead	339,243
Total hours	2,100
Rate	161.54

	Costing #1		Costing #2		Costing #3	
X-Rays	1,018,145	0.303	848,615	0.253	822,829	0.245
Fluoroscopies	1,010,758	0.301	894,596	0.266	877,251	0.261
CAT Scans	661,190	0.197	692,765	0.206	686,347	0.204
MRI Scans	667,906	0.199	922,022	0.275	971,573	0.289
	3,357,999		3,357,999		3,357,999	

The point of this is to show that while the shift from #1 to #2 causes a relatively large change in cost allocation (especially for x-rays and MRIs) the shift from #2 to #3 is more modest, raising the question of whether incurring the incremental costs of #3 are worthwhile.